



Graphing Linear Equations in Slope-Intercept Form

Algebra Worksheet · Grade 8-9

Name: _____

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Learning Objectives

- Identify the slope and y-intercept from an equation in slope-intercept form
- Plot the y-intercept and use rise over run to locate a second point
- Graph linear equations of the form $y = mx + b$ accurately

For each equation, identify the slope and y-intercept, then graph the line by plotting the y-intercept and using the slope to find a second point.

1. Identify the slope and y-intercept of the equation.

$$y = 3x + 2$$

Answer: _____

2. Identify the slope and y-intercept of the equation.

$$y = \frac{1}{2}x - 5$$

Answer: _____

3. Graph the linear equation by finding the y-intercept and a second point using the slope.

$$y = \frac{2}{3}x + 2$$

Answer: _____

4. Graph the linear equation using its slope and y-intercept.

$$y = -\frac{3}{5}x - 3$$

Answer: _____

5. Identify the slope and y-intercept of the equation.

$$y = -\frac{3}{4}x - 6$$

Answer: _____

6. Graph the linear equation using slope-intercept form.

$$y = 2x - 1$$

Answer: _____



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7. Graph the linear equation using slope-intercept form.

$$y = \frac{1}{4}x + 3$$

Answer: _____

8. Rewrite the equation in slope-intercept form and identify the slope and y-intercept.

$$2x + y = 4$$

Answer: _____

9. Graph the linear equation using its slope and y-intercept.

$$y = -x + 5$$

Answer: _____

10. Graph the linear equation using slope-intercept form.

$$y = \frac{3}{2}x - 2$$

Answer: _____



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Remind students that a negative slope means rise goes down (or run goes left) and that the y-intercept is always plotted on the y-axis first.

Solutions

1. Identify the slope and y-intercept of the equation.

$$y = 3x + 2$$

- Compare the equation to $y = mx + b$.
- The coefficient of x is the slope, so $m = 3$.
- The constant term is the y-intercept, so $b = 2$.

Answer: $m = 3, b = 2$

2. Identify the slope and y-intercept of the equation.

$$y = \frac{1}{2}x - 5$$

- Match the equation to the form $y = mx + b$.
- The slope is the coefficient of x , which is $1/2$.
- The y-intercept is the constant term, which is -5 .

Answer: $m = \frac{1}{2}, b = -5$

3. Graph the linear equation by finding the y-intercept and a second point using the slope.

$$y = \frac{2}{3}x + 2$$

- Plot the y-intercept at $(0, 2)$ on the y-axis.
- From $(0, 2)$, use the slope $2/3$ by rising 2 units up and running 3 units right.
- Plot the second point at $(3, 4)$ and draw a straight line through both points.

Answer: y-intercept: $(0, 2)$; second point: $(3, 4)$

4. Graph the linear equation using its slope and y-intercept.

$$y = -\frac{3}{5}x - 3$$

- Plot the y-intercept at $(0, -3)$.
- Because the slope is negative, go down 3 units and right 5 units from the y-intercept.
- Plot the second point at $(5, -6)$ and connect the points with a straight line.

Answer: y-intercept: $(0, -3)$; second point: $(5, -6)$



5. Identify the slope and y-intercept of the equation.

$$y = -\frac{3}{4}x - 6$$

→ Compare the equation to $y = mx + b$.

→ The slope is $-3/4$.

→ The y-intercept is -6 , meaning the line crosses the y-axis at $(0, -6)$.

Answer: $m = -\frac{3}{4}, b = -6$

6. Graph the linear equation using slope-intercept form.

$$y = 2x - 1$$

→ Plot the y-intercept at $(0, -1)$.

→ The slope 2 can be written as $2/1$, so rise 2 units up and run 1 unit right.

→ Plot the second point at $(1, 1)$ and draw the line through both points.

Answer: y-intercept: $(0, -1)$; second point: $(1, 1)$

7. Graph the linear equation using slope-intercept form.

$$y = \frac{1}{4}x + 3$$

→ Plot the y-intercept at $(0, 3)$.

→ Apply the slope $1/4$ by rising 1 unit up and running 4 units right.

→ Mark the point $(4, 4)$ and draw a straight line through both points.

Answer: y-intercept: $(0, 3)$; second point: $(4, 4)$

8. Rewrite the equation in slope-intercept form and identify the slope and y-intercept.

$$2x + y = 4$$

→ Subtract $2x$ from both sides to isolate y .

→ The equation becomes $y = -2x + 4$.

→ The slope is -2 and the y-intercept is 4 .

Answer: $y = -2x + 4; m = -2, b = 4$

9. Graph the linear equation using its slope and y-intercept.

$$y = -x + 5$$

→ Plot the y-intercept at $(0, 5)$.

→ The slope -1 can be written as $-1/1$, so go down 1 unit and right 1 unit.

→ Plot the second point at $(1, 4)$ and draw a line through both points.

Answer: y-intercept: $(0, 5)$; second point: $(1, 4)$

10. Graph the linear equation using slope-intercept form.

$$y = \frac{3}{2}x - 2$$

→ Plot the y-intercept at $(0, -2)$.

→ Use the slope $3/2$ by rising 3 units up and running 2 units right.

→ Plot the second point at $(2, 1)$ and connect the two points with a straight line.

Answer: y-intercept: $(0, -2)$; second point: $(2, 1)$



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